

DETERMINISTIC PDA

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AP/CSE

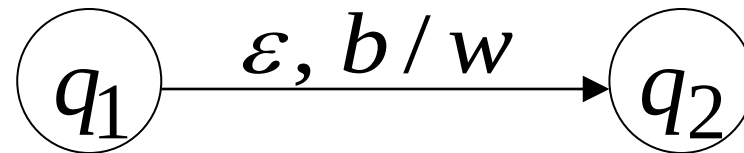
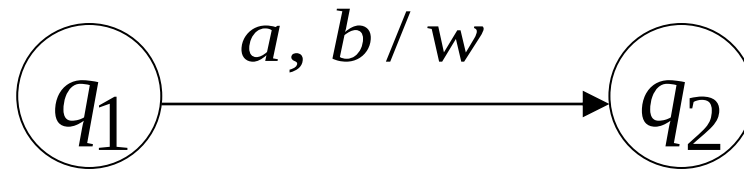
LEARNING OBJECTIVE

- To Design pushdown automata for any CFL (K3)
 - To Understand the concept of Deterministic PDA

DETERMINISTIC PDA'S

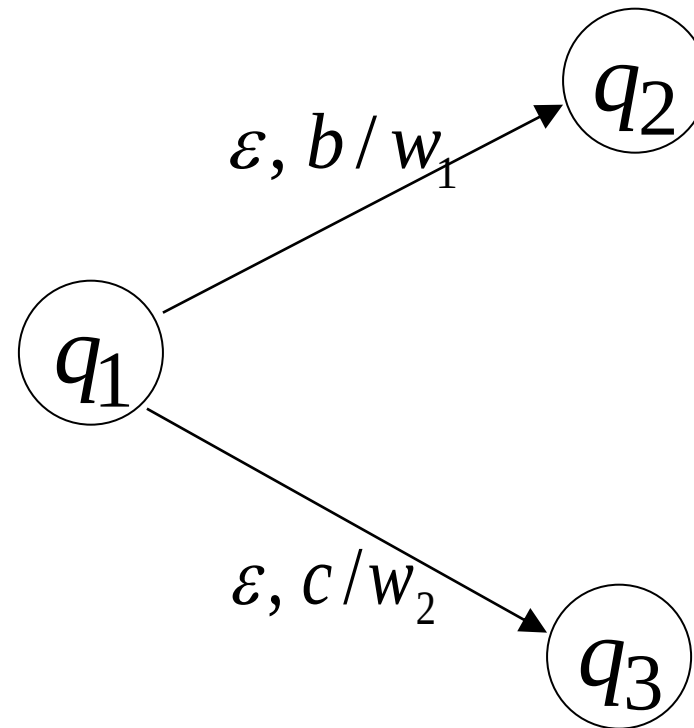
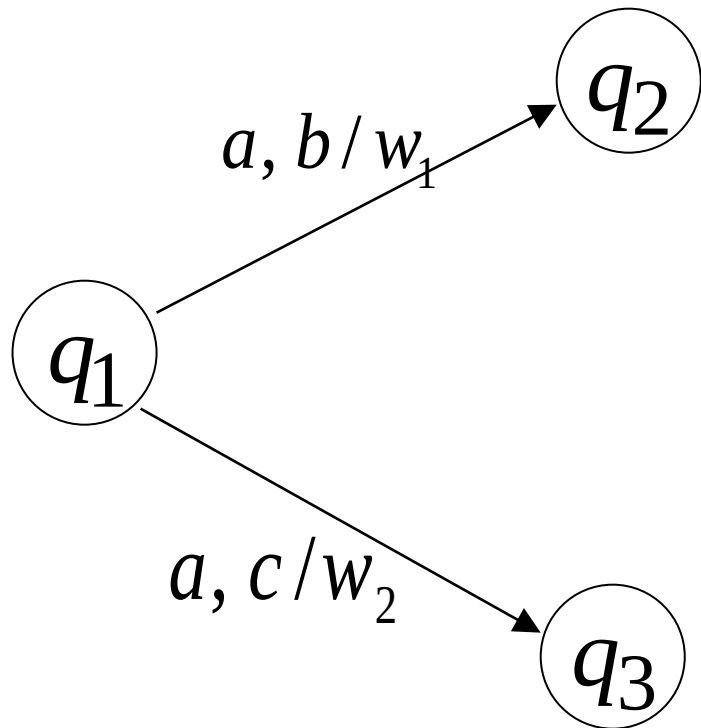
- A PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ is said to be deterministic if
 - $\delta(p, a, \beta) = (q, \gamma)$
 - ie. To be deterministic, **there must be at most one choice** of move for any state p , input symbol a , and stack symbol β .
 - $\delta(p, \varepsilon, \beta)$ is not empty then $\delta(p, a, \beta)$ must be empty for every $a \in \Sigma, p \in Q, \beta \in \Gamma$.
 - ie. **there must not be a choice between using input ε or real input.**
 - Formally, $\delta(p, \varepsilon, \beta)$ and $\delta(p, a, \beta)$ cannot both be nonempty.

ALLOWED TRANSITIONS



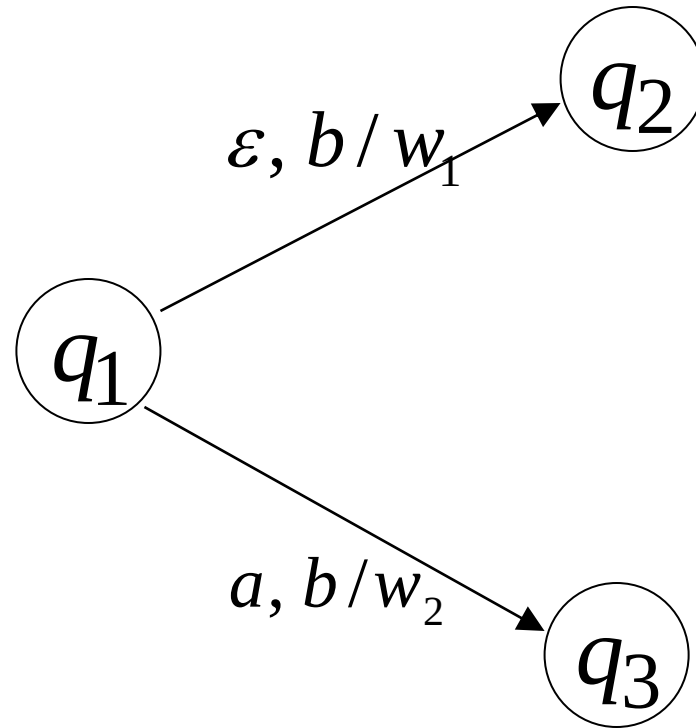
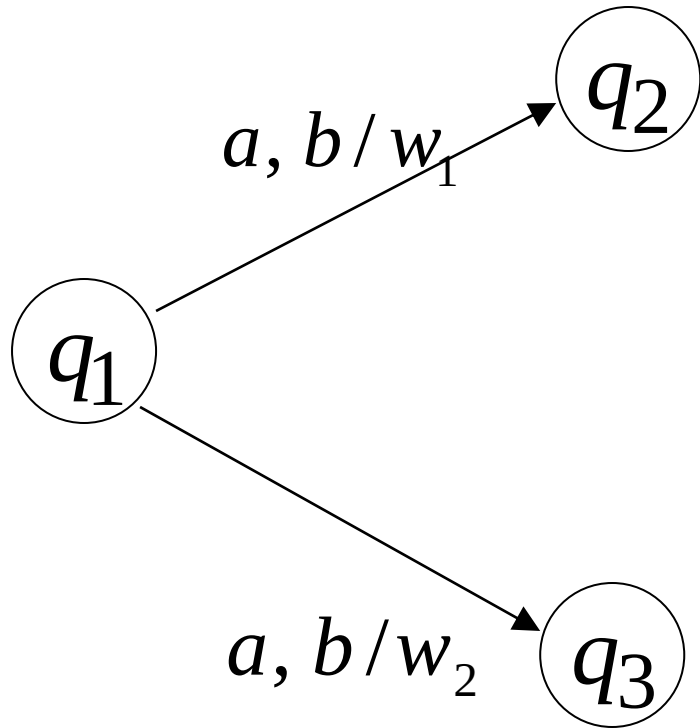
(deterministic choices)

ALLOWED TRANSITIONS



(deterministic choices)

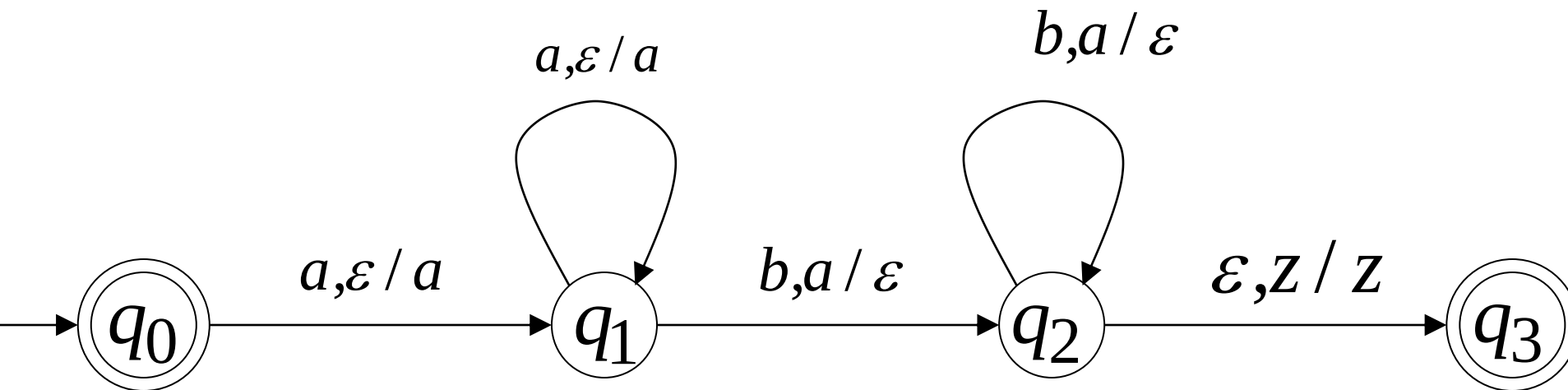
NOT ALLOWED TRANSITIONS



(non deterministic choices)

EXAMPLE

$$L(M) = \{a^n b^n : n \geq 0\}$$



DETERMINISTIC CF

- A language L is Deterministic CF if there exists some DPDA that accepts it.

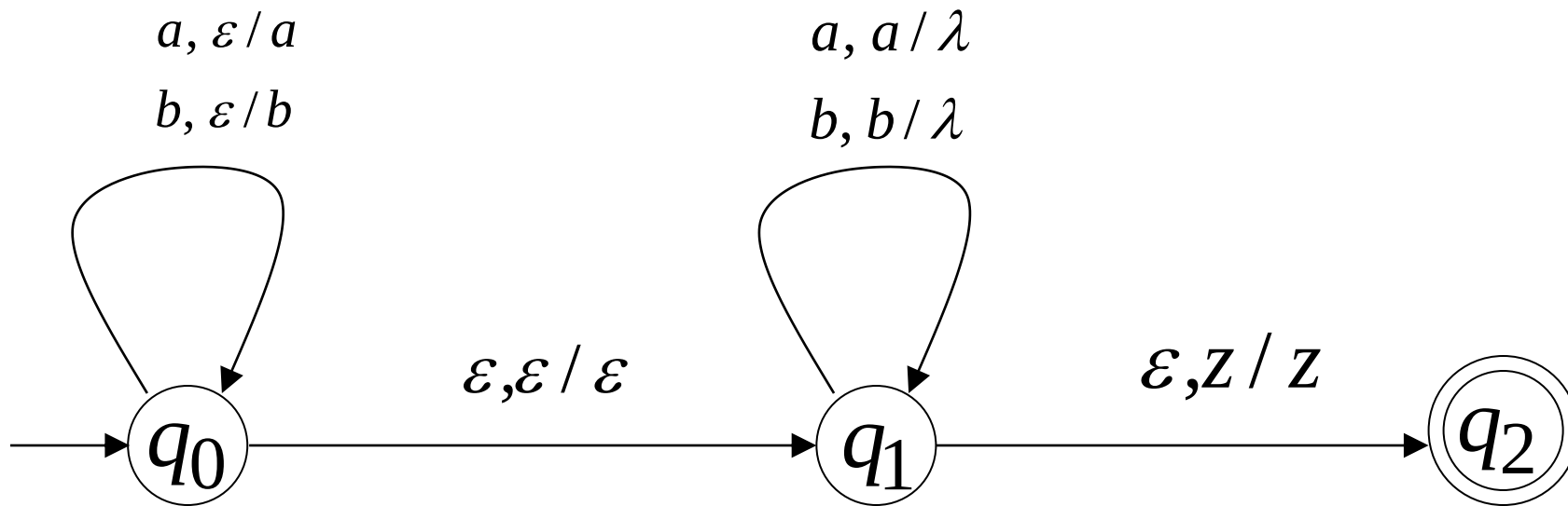
Example:

The language $L(M) = \{a^n b^n : n \geq 0\}$

is **deterministic context-free**

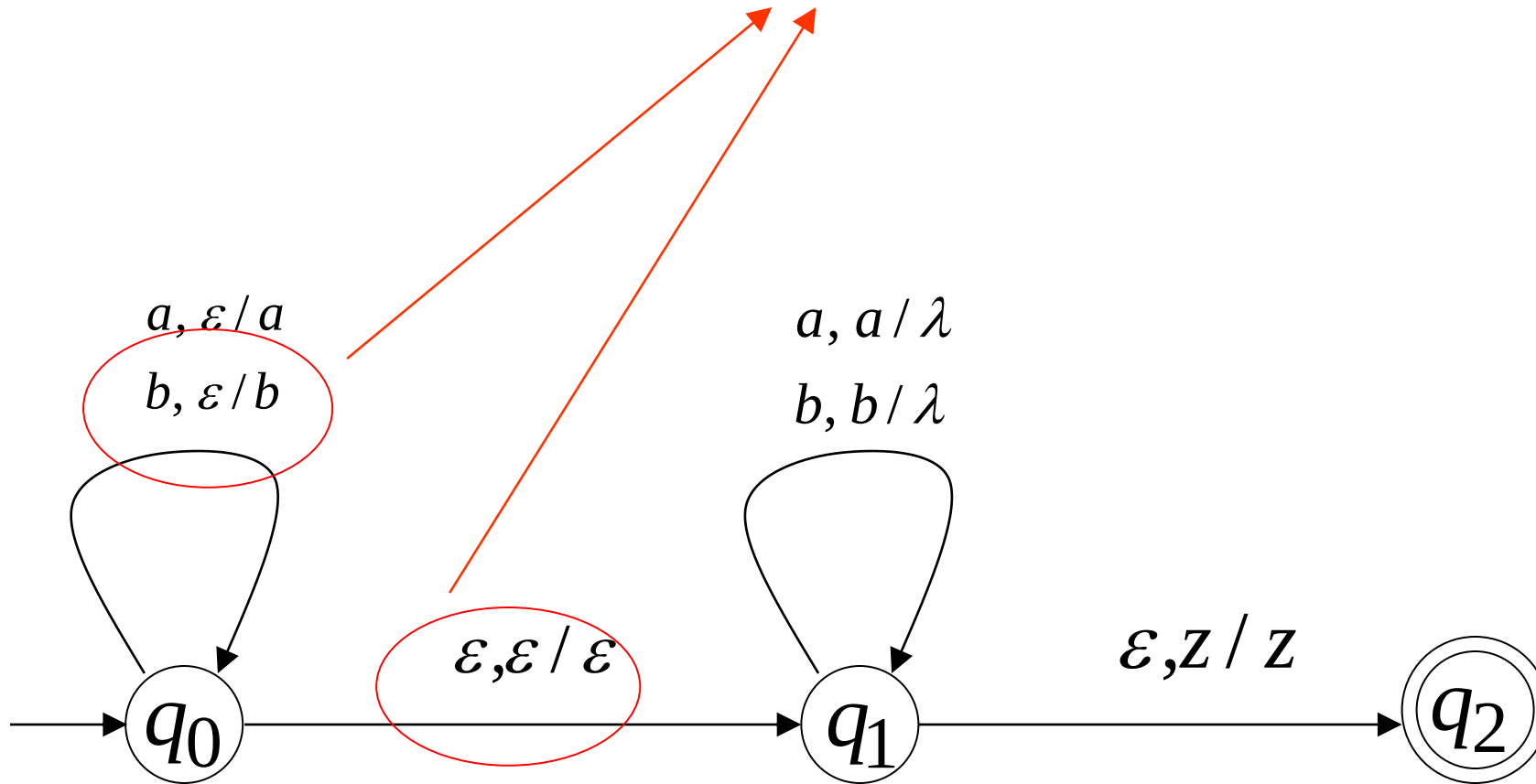
EXAMPLE OF NON-DPDA (PDA)

$$L(M) = \{vv^R : v \in \{a,b\}^*\}$$



EXAMPLE OF NON-DPDA (PDA)

Not allowed in DPDAs



SUMMARY

- Definition of Deterministic PDA

TEST YOUR KNOWLEDGE

- With reference of a DPDA, which among the following do we perform from the start state with an empty stack?
 - a) process the whole string
 - b) end in final state
 - c) end with an empty stack
 - d) all of the mentioned

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008